Nuclear Energy: Fission and Fusion

Introduction to Nuclear Energy

Nuclear energy is the energy stored in the nucleus, the core, of an atom. It is released when the nucleus is split or when light nuclei are combined. The enormous energy comes from the conversion of a small amount of mass into energy, as described by Albert Einstein's equation E = mc squared, where E is energy, m is mass and c is the speed of light. One kilogram of uranium-235 can release millions of times more energy than one kilogram of coal. Nuclear energy is used to produce electricity in nuclear power plants, to power submarines and ships, in medicine and in weapons. Nuclear power produces almost no greenhouse gases during operation, which makes it important for climate goals.

History

Radioactivity was discovered by Henri Becquerel in 1896, and Marie Curie and Pierre Curie studied it. In December 1938 the German scientists Otto Hahn and Fritz Strassmann discovered nuclear fission, which Lise Meitner and Otto Frisch explained. On 2 December 1942 Enrico Fermi's team created the first controlled nuclear chain reaction at the University of Chicago. The first nuclear power plant to supply electricity to a grid began in Obninsk, Soviet Union, in 1954. Major accidents occurred at Chernobyl in 1986 and Fukushima in 2011, which led to stronger safety rules.

What Is Nuclear Fission?

Nuclear fission is the process in which a heavy nucleus, such as uranium-235 or plutonium-239, is split into two lighter nuclei when it is hit by a neutron. The split releases a large amount of energy, plus two or three additional neutrons. These neutrons can strike other uranium nuclei and split them, leading to a chain reaction. In a nuclear reactor, the chain reaction is controlled by control rods, made of materials such as boron or cadmium that absorb neutrons, and by a moderator, such as water, heavy water or graphite, that slows neutrons down. The heat generated boils water into steam, which spins a turbine connected to a generator. Fission is used in all of today's nuclear power plants.

What Is Nuclear Fusion?

Nuclear fusion is the process in which two light nuclei, such as the hydrogen isotopes deuterium and tritium, combine to form a heavier nucleus, helium, releasing a huge amount of energy. Fusion is what powers the Sun and other stars. It needs extremely high temperatures, over 100 million degrees Celsius, and high pressure so that the positively charged nuclei can overcome their electric repulsion. Fusion releases about three to four times more energy per kilogram of fuel than fission, produces little long-lived radioactive waste and has abundant fuel, since deuterium can be extracted from seawater. Fusion is still experimental. The ITER project in southern France, in which India is a partner, is building a large tokamak device to demonstrate it. In December 2022 the National Ignition Facility in the United States achieved fusion ignition, producing more energy from the fusion reaction than the laser energy delivered to the fuel, though not more than the electricity used to run the lasers.

Difference Between Fission and Fusion

Fission splits a heavy nucleus into smaller ones, while fusion joins light nuclei into a heavier one. Fission works at moderate temperatures and is used in reactors today; fusion needs extreme temperatures and is not yet commercial. Fission produces radioactive waste that lasts for thousands of years; fusion produces much less. Fission can be sustained as a controlled chain reaction; the Sun is a natural fusion reactor.

Which Fuel Is Used in Most Nuclear Reactors?

Most nuclear reactors use uranium as fuel, specifically the isotope uranium-235. Natural uranium contains only about 0.7 percent of uranium-235 and the rest is uranium-238, so for most reactors the uranium is enriched to about 3 to 5 percent U-235 and formed into ceramic pellets of uranium dioxide inside metal fuel rods. Indian pressurised heavy water reactors (PHWRs) can run on natural uranium because they use heavy water as the moderator. Plutonium-239 is produced in reactors from uranium-238 and can be used as fuel. India has large reserves of thorium, and Dr. Homi Bhabha designed a three-stage nuclear power programme to eventually use thorium: first PHWRs, then fast breeder reactors using plutonium, and finally thorium-based reactors.

Nuclear Power in India and Tamil Nadu

India's nuclear power plants are operated by the Nuclear Power Corporation of India Limited (NPCIL) and supply about 3 percent of the country's electricity. India's first nuclear power station was at Tarapur in Maharashtra, which began operating in 1969. In Tamil Nadu, the Kudankulam Nuclear Power Plant in Tirunelveli district, built with Russian assistance, has Russian-designed VVER-1000 reactors, and its units 1 and 2 are operating, with more units under construction. The Madras Atomic Power Station at Kalpakkam, near Chennai, has two 220-megawatt PHWR units operating since 1984 and 1986. Kalpakkam also hosts the Indira Gandhi Centre for Atomic Research and a 500 MWe Prototype Fast Breeder Reactor nearing commissioning. India aims to greatly expand nuclear capacity by 2047, including small modular reactors.

Challenges and Future

Challenges include radioactive waste disposal, safety, high initial cost, public acceptance, and the risk of nuclear weapon proliferation. The future includes small modular reactors, thorium reactors and, in the longer term, commercial fusion power.

Quick Facts

- Fission splits heavy nuclei such as uranium-235; fusion joins light nuclei such as deuterium and tritium and powers the Sun.
- Most reactors use enriched uranium-235; Indian PHWRs use natural uranium with heavy water.
- Kudankulam and Kalpakkam (Madras Atomic Power Station) are nuclear sites in Tamil Nadu.
- The first controlled chain reaction was achieved by Enrico Fermi on 2 December 1942.
- India's three-stage nuclear programme was designed by Homi Bhabha to use thorium.